FLUID MECHANICS AND HYDRAULIC MACHINES LAB

II B.TECH II Semester: MECHANICALENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6ME17	PCC	0	0	2	1	40	60	100

COURSE OVERVIEW :

This course will provide a basic understanding of calibration of flow measuring devices such as venturi meter, orifice meter, notches, water meter, rotameter etc used in channels and pipes and Losses in pipes by major and minor loss apparatus. Measurements of critical Performance parameters like efficiency, flow rate etc of various hydraulic machines such as turbines, pumps are also measured in this lab. Students can perform tests and find the flow rate of equipment like venturimeter, orifice meter, and notches and can calibrate them. Darcy's as well as Chezy's coefficient of friction for different pipes can be found out in pipe friction apparatus. Also students can find out the performance of centrifugal and reciprocating pumps.

COURSE OUTCOMES:

At the end of course students are able to

1. Develop procedure for standardization of experiments.
2. Calibrate flow discharge measuring devices used in pipes.
3. Determine the major and minor losses in a given pipe.
4. Prove that the total head at any point along the fluid flow is same.
5. Test the performance of pumps and turbines.

LIST OF EXPERIMENTS:

1. Impact of jets on Vanes.
2. Performance Test on Pelton Wheel.
3. Performance Test on Francis Turbine.
4. Performance Test on Kaplan Turbine.
5. Performance Test on Single Stage Centrifugal Pump.
6. Performance Test on Multi Stage Centrifugal Pump.
7. Performance Test on Reciprocating Pump.
8. Calibration of Venturimeter.
9. Calibration of Orifice meter.
10. Determination of friction factor for a given pipe line.
11. Determination of loss of head due to sudden contraction in a pipeline.